COLLEGE OF ENGINEERING & TECHNOLOGY
Course:Computer Graphics
Course No. CC416 Term : Fall 2023-2024
Name : ....
Class : ....

Model: B

Questions for the 7th Week Examination

Part I: (20 points)

Question One (8 Points)

a) Complete the following questions:

1) .... is the number of *gray levels* per unit measure of image amplitude (often power of 2).
2) The Digital graphics input devices such as .... and ....
3) In Windows XP the .... save its images automatically in bitmap format,
4) if the phosphor persistence is too .... then the image may still appear blurry
5) ________................................ is an example of raster displays
6) The persistence of a phosphor is defined as the time it takes for emitted light to decay to ....
7) A gamer is playing a fast-paced game on a CRT monitor. The game requires the monitor to display images with minimal motion blur. The monitor has a refresh rate of 60 Hz and a phosphor persistence of 25 ms. refresh rate fast enough............................ to eliminate motion blur

b) **Select one of the choices that best suites the given statement**

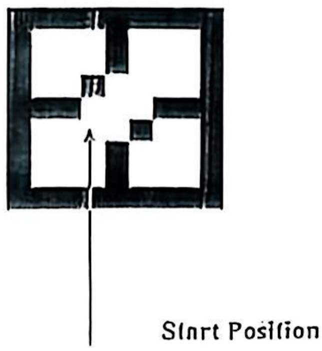
1) Interactive computer graphics uses various kind of input devices such as
   a) Mouse b) Graphic tablet c) Joystick d) a, b, and c
2) The .... line drawing algorithm is a time consuming method
   a) DDA b) Bresenham's c) Direct Scan conversion
3) .... used in the Unix world for program icons and background images
   a)PNG b) JPG c) XBM and XPM
4) A 16*16 array of black and white pixels could be represented by ....
   a) 64 bytes b) 32 bytes c) 128 bytes d) 96 bytes
5) RGB true color model has _______ color depth
   a) 24bit b) 32bit c) 64bit
6) Intensity range of 4-bit pixel image is ....
   a) 0 to 5 b) 0 to 63 c) 0 to 15 d) 0 to 127
7) Two basic techniques for producing color display with a CRT are ....
   a) Shadow mask and random scan b) Beam penetration and shadow mask method
   c) Random scan and raster scan d) None of above
8) LCD are commonly used in
   a) Calculators b) Portable c) Laptop computers d) All of these

Question Two (4 Points)

<u>Apply</u> *decision parameters* for generating points along a **<u>line path</u>** through point: (2,9) and second point (5,14) using ***Bresenham algorithm***.

Question Three (3 Points)

Apply the 8-connected Flood Filling Algorithm to fill the polygon shown in the following image segment starting from the labeled pixel s



Question Four (5 Points)

Given a circle whose center is at *(3, 4)* and radius *r =7 pixels*, **demonstrate** the ***midpoint circle algorithm*** to draw the circle

Hints:

$p_0 = 1 - r$

If $(p_k < 0)$ $\text{Plot}(x_k + 1, y_k)$

$$p_{k+1} = p_k + 2(x_k + 1) + 1$$

Otherwise

$$\text{Plot}(x_k + 1, y_k - 1)$$

$$p_{k+1} = p_k + 2(x_k + 1) + 1 - 2(y_k - 1)$$

Best Wishes

COLLEGE OF ENGINEERING & TECHNOLOGY
Course:Computer Graphics
Course No : CC416 Term : Fall 2023-2024
Name : ....
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Model : A

Questions for the 7th Week Examination

Part I: (20 points)

Question One (8 Points)

a) Complete the following questions:

1) .... is the number of image points per unit measure in the image domain
2) The analog graphics input devices devices such as .... and ....
3) In Windows XP the Paint program save its images automatically in .... format,
4) if the refresh rate is too low, then the image may ....
5) In a raster-scan system, the electron beam is swept across the screen, one row at a time, from top to bottom. Each row is referred to as a ....

b) Select one of the choices that best suites the given statement

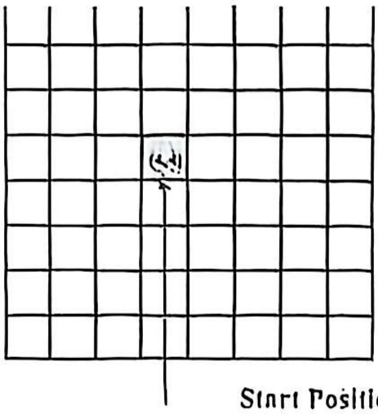
1) The number of pixels stored in the frame buffer of a graphics system is known as
   a) Resolution b) Depth c) Resalution
2) The .... line drawing algorithm is good for hardware implementation
   a) DDA b) Bresenham's c) Direct Scan conversion
3) Raster scan is .... expensive than random scan
   a) More b) Less c. c) Both a & b d) None
4) .... is mostly used for loading the fast web pages. It also help to makes great banner and logo for different webpage
   a)GIF b) BMP c) TIFF
5) Consider an image of size M X N with 64 gray levels. The total number of bits required to store this digitized image is ....
   a) M X N X 63 b) M X N X 64 c) M X N X 6 d) M X N X 8
6) If we use direct coding of RGB values with 6 bits each for red and blue, and 4 bits for green . the number of possible simultaneous colors are ....
   a) 2^{16} b) 2^{10} c) 2^{24} d) None of these
7) Lower persistence phosphorus needs .... refresh rate
   a) Lower b) Higher c) Medium d) None of these
8) Random scan systems are designed for
   a) Line drawing application b) Pixel drawing application
   c) Color drawing application d) None of these
9) The return to the left of the screen, after refreshing each scan line is called..........
   a) Horizontal retrace b) Vertical retrace c) Refresh rate
10) A phosphor has a persistence of 8 ms. What is the minimum refresh rate required to avoid noticeable flickering?
   a) 50 Hz b) 60 Hz c.)75 Hz d) 100 Hz

Question Two (4 Points)

Apply *decision parameters* for generating points along a line path through point: (5,3) and second point (1,7) using *Bresenham algorithm.*

Question Three (3 Points)

Apply, *step by step*, the span flood filling algorithm to fill regions with boundaries defined in the following image segment?



Question Four (5 Points)

Given a circle whose center is at *(3, 3)* and radius $r = 5$ *pixels*, demonstrate the *midpoint circle algorithm* to draw the circle.

Hints:

$p_0 = 1 - r$

If $(p_k < 0)$ Plot$(x_k + 1, y_k)$

$$p_{k+1} = p_k + 2(x_k + 1) + 1$$

Otherwise

Plot$(x_k + 1, y_k - 1)$

$$p_{k+1} = p_k + 2(x_k + 1) + 1 - 2(y_k - 1)$$